

Standard-Cell Characterization

Methods for sg13g2_stdcell_hv



fo4.py (reference) · CharLib 2.1.0 · lctime 0.0.26 — the measurement physics behind the Liberty file

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1 Scope and headline result

Three tools touched the sg13g2_stdcell_hv Liberty file, in three different roles:

- **work/fo4.py** — the *reference anchor*: a 96-line ngspice deck that measures the FO4 delay and the input capacitance of the thick-oxide inv_1 against the thin-oxide original. Its two ratios (2.66× delay, 2.20× capacitance) set the slew and load grids of the shipped library, and its rail-biased AC capacitance (5.87 fF) is the reference every other capacitance number in this report is judged against.
- **CharLib 2.1.0** (commit 6859faf, driven through the infinitymdm PySpice 1.6 fork) — the *production characterizer* that generated the shipped NLDM tables, configured with the charge_integration input- capacitance procedure and, for the sequential cells, the project's own seq_delay_procedure.py registered through CharLib's procedure registry.
- **lctime 0.0.26** (LibreCell) — the *independent cross-check*: work/lctime_compare.py re-characterized eight combinational cells on identical grids and models and aligned 3 132 table points against the shipped library.

Every method claim below was verified in the installed source of the three tools (file and line references throughout; a provenance appendix closes the report). The comparison in one table, before the derivations:

quantity	fo4.py (reference)	CharLib 2.1.0 as configured	lctime 0.0.26
delay / slew load	four real inverter gates	lumped capacitor	lumped capacitor
stimulus ramp	shaped by a driver stage	PWL, stretched by 1/0.6 (Liberty-correct)	PWL equal to the raw slew number (convention error)
extraction	ngspice .meas trig/ targ	ngspice .meas trig/ targ	numpy interpolation of waveforms
input capacitance			

quantity	fo4.py (reference)	CharLib 2.1.0 as configured	lctime 0.0.26
	AC at 1 MHz, rail-biased, mean of both rails	charge integration, worst edge	constant-current slope 10 μ A, 20–80 % window
inv_1 C_{in} result	5.87 fF	6.44 fF (+9.7 %)	8.92 fF (+52 %)
internal power	not measured	not implemented in CharLib 2.1.0	rectangle-rule energy integration (load energy not subtracted)
leakage	not measured	DC operating point per input state, all 2^N states	unimplemented stub
setup / hold	—	pass/fail bisection, 1.5 $\times$ C2Q degradation (local procedure)	Brent root-finding, 10 ps absolute pushout
clock $\rightarrow$ Q	—	local procedure (upstream is a stub)	implemented; constraint search at zero output load
simulator coupling	ngspice -b batch	libngspice in-process (shared)	ngspice subprocess, ASCII wrdata

Answer to the standing question — can lctime use CharLib’s charge-integration method? Not as shipped: lctime 0.0.26 contains no code path that integrates a current anywhere (no trapz, no cumsum, no ngspice integ in any generated .control block). Its input capacitance is a constant-current secant slope, which is where its +52 % error comes from. However, lctime’s simulator coupling *already* returns branch currents with the time vector — it integrates supply and gate current for its internal-power tables — so a charge-integration capacitance is a localized, three-part patch to one function. Section 8 gives the exact changes.

2 What the Liberty NLDM asks for

All three tools ultimately serve the same data model, so the definitions come first. The library header fixes eight trip points; the shipped file uses the industry-standard set

$$V_{th,in} = V_{th,out} = 0.5 V_{DD}, \quad V_{slew,lo} = 0.2 V_{DD}, \quad V_{slew,hi} = 0.8 V_{DD}$$

and defines, per timing arc, the propagation delay and the output transition time as threshold-crossing intervals of the simulated waveforms:

$$t_{pd} = t(v_{out} = 0.5 V_{DD}) - t(v_{in} = 0.5 V_{DD})$$

$$t_{tran} = t(v_{out} = 0.8 V_{DD}) - t(v_{out} = 0.2 V_{DD})$$

(rising conventions shown; falling edges mirror the thresholds). The NLDM `cell_rise / rise_transition` groups tabulate these two quantities as functions of two independent variables,

$$t_{pd} = f(s_{in}, C_L), \quad t_{tran} = g(s_{in}, C_L),$$

where s_{in} is the *input* transition time measured between the same 20 %/80 % slew thresholds, and C_L the lumped load. The shipped library uses 7×7 grids per arc; the grid values are the thin-oxide library's grids scaled by the fo4.py ratios so the tables span the same electrical territory. Everything a characterizer does for a combinational arc reduces to filling f and g point by point from transient simulations, and every methodological difference between the three tools is a difference in *stimulus construction*, *load modelling*, or *waveform extraction* for those same two equations.

A physical scale for the delays themselves comes from logical effort: with $\tau = R_{inv}C_{in}$ the technology time constant, a stage driving an electrical effort $h = C_{out}/C_{in}$ has normalized delay

$$d = gh + p,$$

and the fanout-of-4 inverter delay ($g = 1$, $h = 4$) is the canonical technology speed metric fo4.py measures directly: $t_{FO4} = \tau(4 + p)$. It is a *ratio-preserving* metric — sizing every cell by the same factors leaves g and p nearly unchanged — which is exactly why one FO4 number per library (142.4 ps HV vs 53.6 ps LV) legitimately rescales an entire grid set.

3 Input capacitance: three answers to one question

3.1 The physics being sampled

The input pin of a CMOS gate is not a capacitor; it is a nonlinear, *transcapacitive* charge reservoir. The gate charge is a function of the gate voltage **and** of every other terminal voltage, so the differential capacitance seen while the input traverses its switching trajectory is

$$C_{in}(v) = \frac{dQ_G}{dv_{in}} = C_{gs}(v) + C_{gb}(v) + C_{gd}(v)\left(1 - \frac{dv_{out}}{dv_{in}}\right).$$

The last term is the Miller effect: while the cell switches, the output moves *against* the input with the incremental gain $A_v = dv_{out}/dv_{in} < 0$, so C_{gd} contributes $C_{gd}(1 + |A_v|)$ — a sharp peak centred on the switching threshold V_M , where $|A_v|$ is largest. Away from V_M , with the output pinned at a rail, $A_v \approx 0$ and C_{in} relaxes to its small end-point values.

Every “input capacitance” is therefore a *functional* of $C_{in}(v)$ — a weighted average over some window of the input swing — and the three tools choose three different windows. That single observation explains all three numbers.

3.2 fo4.py: small-signal AC at the rails

fo4.py drives an isolated inverter input with a 1 V AC source at 1 MHz, once biased at $v = 0$ and once at $v = V_{DD}$ (fo4.py:38–55). For a linear one-port $Y(j\omega) = G + j\omega C$, the measured current at unit drive gives

$$C_{AC} = \frac{\text{Im}\{I\}}{2\pi f|V|} \quad (\text{evaluated at } f = 1 \text{ MHz}).$$

The frequency is chosen so that gate leakage is invisible ($G \ll \omega C$: at 1 MHz, $\omega C \approx 3.7 \times 10^{-8}$ S for 5.9 fF, orders above any gate conductance) yet low enough that non-quasi-static channel effects play no role. The deck comments state the bias rationale explicitly: at either rail the inverter sits in a zero-gain region, so $A_v \approx 0$ and the Miller term vanishes —

$$C_{AC} = \frac{1}{2}[C_{in}(0) + C_{in}(V_{DD})] \quad (\text{endpoint sample, no Miller}).$$

A mid-rail bias would instead sample the peak of $C_{in}(v)$ and Miller-multiply C_{gd} into the answer. Result: **5.87 fF** for sg13g2_hv_inv_1.

3.3 CharLib: charge integration

The configured procedure (charlib/characterizer/procedures/pin_capacitance/charge_integration.py) drives the pin with a PWL *voltage* ramp — VSS up to VDD, wait, back down — and lets ngspice integrate the stimulus-source branch current over each edge window:

```
.meas TRAN q_rise integ i(vstim) from=... to=...
.meas TRAN q_fall integ i(vstim) from=... to=...
```

then applies the defining relation of effective capacitance (charge_integration.py:100–135):

$$C_{int} = \frac{|\Delta Q|}{\Delta V} = \frac{1}{V_{DD}} \int_{\text{edge}} i_{in}(t) dt = \frac{1}{V_{DD}} \int_0^{V_{DD}} C_{in}(v) dv.$$

This is the *full-swing mean* of $C_{in}(v)$ along the real switching trajectory: the Miller charge $Q_M \approx C_{gd}(\Delta v_{in} + \Delta v_{out})$ is included in ΔQ , but it is spread over the full denominator V_{DD} . Non-driven pins are isolated with $10 \text{ G}\Omega \parallel 1 \text{ pF}$; the ramp time defaults to the fastest grid slew and the wait time to 1000× that, so each edge integrates to completion. The reported capacitance is the worst (larger) of the two edges. Result: **6.44 fF** (+9.7 % vs the AC reference — that surplus *is* the Miller and nonlinearity charge, not an error; it is the charge a driving cell genuinely must deliver across a full transition).

Worth recording: CharLib's *upstream default* is a different procedure, `ac_sweep`, which fits capacitance as the slope of the admittance *magnitude* versus frequency, $C = d|Y|/df$. Since $|Y| = 2\pi fC$ for a capacitor, the correct estimator is $C = \frac{1}{2\pi} d|Y|/df$; the missing $1/2\pi$ makes the default overestimate by $\approx 6.28\times$, and the observed 6.8–7.5× (inv_1: 43.8 fF) is exactly 2π

compounded with the Miller contribution of its mid-transition drive. This is why the project configuration selects `charge_integration`.

3.4 lctime: constant-current secant slope

`lctime` does something else entirely (`lctime/characterization/input_capacitance.py`, docstring: “*Measurement of the input capacitance by driving the input pin with a constant current*”). A fixed source $I = 10\ \mu\text{A}$ (hard-coded, `util.py:162`) charges the pin; the pin *voltage* alone is recorded; and the capacitance is the secant slope between the two slew trip points (`input_capacitance.py:273–297`):

$$C_{\text{slope}} = \frac{I}{\Delta V / \Delta t} = \frac{I[t(0.8V_{DD}) - t(0.2V_{DD})]}{0.6V_{DD}}.$$

From first principles this is again a windowed mean of the same $C_{in}(v)$: with $C_{in}(v) dv/dt = I$,

$$\Delta t = \frac{1}{I} \int_{0.2V_{DD}}^{0.8V_{DD}} C_{in}(v) dv \Rightarrow C_{\text{slope}} = \frac{1}{0.6V_{DD}} \int_{0.2V_{DD}}^{0.8V_{DD}} C_{in}(v) dv.$$

The window $[0.2, 0.8]V_{DD}$ *contains* the Miller peak at V_M and *excludes* the low-capacitance tails near the rails; the Miller charge is divided by $0.6V_{DD}$ instead of V_{DD} . Both effects push the same direction, which orders the three estimators from first principles alone — the Miller-free endpoint sample below the full-swing mean below the peak-window mean:

$$C_{AC} < C_{int} < C_{\text{slope}}$$

— and the measurements land in exactly that order: $5.87 < 6.44 < 8.92\ \text{fF}$, the slope method **+52 %** above the reference. `lctime` averages the result over all 2^n static states of the other inputs and over both edge directions (reduction per calc-mode: mean for typical), and carries a genuine bug while doing so: the returned dictionary swaps the labels, assigning the falling measurement to `rise_capacitance` and vice versa (`input_capacitance.py:318–321`). The default capacitance attribute, being the mean of both, hides the swap.

4 Delay and output slew

4.1 Stimulus: the slew convention is where the tools diverge

Liberty defines s_{in} between the 20 %/80 % thresholds. A characterizer that drives the pin with a single linear ramp of total duration T ($0 \rightarrow 100\%$) realizes a Liberty slew of

$$s = (0.8 - 0.2)T = 0.6T \quad \Longleftrightarrow \quad T = \frac{s}{0.6}.$$

CharLib gets this right. Its `slew_pwl` helper (`utils.py:49–65`, explicitly citing the Liberty User Guide) stretches the requested slew to the full ramp, $T = s/(V_{hi} - V_{lo}) = s/0.6$, before building the PWL source.

lctime does not. Its StepWave for combinational stimulus is built with `rise_threshold=0`, `fall_threshold=1` (timing_combinatorial.py:186–191), i.e. the table’s slew number is used directly as the 0→100 % ramp time. The circuit therefore sees an edge whose *Liberty* slew is only 0.6 s — every lctime table point at index *s* is actually a measurement at 0.6 s. Since $\partial t_{pd}/\partial s_{in} > 0$, lctime under-reads delays, negligibly at fast edges and severely at slow ones. The cross-check quantifies it: over the STA-relevant region the two tools agree to a median 2.9 % on delays and 0.0 % on output transitions (3 132 points, 8 cells), but at the 3.36 ns slew point of `inv_1` (0.396 pF) a direct ngspice measurement gives 2.3605 ns where CharLib’s table says 2.3571 ns (**+0.15 %**) and lctime says 1.9158 ns (**–19 %**); re-interpolating the CharLib table at 0.6 s reproduces lctime to 0.25 %, nailing the mechanism. Ironically lctime *reads* all eight trip points from the Liberty template header (util.py:377–398) and measures with them correctly — only the stimulus construction ignores them.

fo4.py sidesteps the question: its device under test is driven by a real inverter stage (`Xdrv`), so the edge shape is whatever the technology produces, and the measured 50 % crossings use ngspice `.meas` with `rise=2/fall=2` — the *second* edge, past the initial-condition transient, so the measurement is taken in periodic steady state.

4.2 Load modelling

Both characterizers drive a lumped capacitor: C_L to ground on the output (CharLib `delay.py`: 83; lctime deck `Cload` elements). `fo4.py` loads the DUT with **four actual inverter inputs** — nonlinear $C_{in}(v)$ loads that also kick Miller charge back into the driving node. A lumped capacitor equal to $4 C_{in}$ is the NLDM abstraction of that load; the FO4 deck is the ground truth the abstraction is checked against. This is precisely why the library’s load grid is anchored to a *measured* C_{in} ratio rather than a nominal one.

Side-input handling differs in a way that shows up in multi-input cells: CharLib simulates every non-masking static state of the other inputs and records the **worst case**, `criterion = max` (`delay.py`:16, 188–190); lctime enumerates the unateness-consistent states and reduces with the `calc-mode` function — `np.mean` under the default `typical` mode (timing_combinatorial.py:77–81). Two tables built from identical simulations can therefore legitimately differ on any cell where the arc delay depends on the side-input state (stack position effects).

4.3 Extraction

`fo4.py` and CharLib let ngspice do the measurement — `.meas tran ... trig v(in) val=... targ v(out) val=...` — which interpolates crossings inside the simulator on the adaptive time grid. lctime post-processes: it normalizes the stored waveforms to $[0, 1]$, mirrors falling edges, and linearly interpolates the bracketing samples of each threshold crossing in numpy (util.py:251–374). Both are first-order interpolations of the same transient; the difference is not accuracy but *plumbing* — and the plumbing matters, as §7 shows: PySpice’s subprocess backend silently drops `.meas` results, which is why CharLib must run in shared-library mode, while lctime never uses `.meas` at all and is immune.

5 Energy, power, leakage

5.1 The energy bookkeeping a Liberty file expects

Per rising output transition the supply delivers charge $Q = \int i_{VDD} dt$, hence energy

$$E_{supply} = V_{DD} \int i_{VDD}(t) dt = V_{DD} \Delta Q.$$

Of this, $C_L V_{DD}^2$ is associated with the external load (half stored on C_L , half dissipated in the pull-up network), and the remainder is *internal*: parasitic-node charging plus the short-circuit (crowbar) charge that flows while both networks conduct near V_M . Liberty's `internal_power` tables are meant to carry only that remainder, because the STA tool separately computes the switching energy of the external net from its own capacitance:

$$E_{int} = V_{DD} \int i_{VDD} dt - C_L V_{DD}^2 \quad (\text{rising edge}).$$

CharLib 2.1.0 does not measure internal power at all. The package contains no procedure writing `internal_power`, `rise_power` or `fall_power` — energy appears only as a unit definition — and `Characterize.analyse_cell` schedules exactly: pin capacitance, then delay + leakage (combinational) or the constraint procedures (sequential). The shipped Liberty consequently has no internal-power groups; any power-aware flow consuming it sees switching and leakage power only.

lctime is the only tool of the three that writes power tables, inside the same transient as the delay measurement (`timing_combinatorial.py:263–274`):

```
gate_energy    = np.mean(gate_current * input_voltage) * dt
supply_energy  = np.mean(supply_current * vdd) * dt
switching_energy = gate_energy + supply_energy
```

Two defects are visible from first principles. First, the quadrature: $\text{mean}(f) \cdot (t_N - t_0)$ equals $\int f dt$ **only on a uniform time grid**; ngspice's adaptive stepping clusters points around the edges, where $i v$ is largest, so the sample mean over-weights the transition region. The unbiased estimator on the same data is the trapezoidal sum

$\int f dt \approx \sum_k \frac{1}{2}(f_k + f_{k+1})(t_{k+1} - t_k)$, i.e. `np.trapz(f, time)` — available but unused. Second, the accounting: no $C_L V_{DD}^2$ subtraction is performed, so the tabulated “internal” energy includes the full load-charging energy and grows linearly with C_L ; an STA tool then counts that energy a second time as switching power. Sequential cells get no power tables even from `lctime` (the supply current is retrieved in the FF harness but never integrated).

5.2 Leakage

Static power is a per-state quantity because subthreshold leakage depends exponentially on the bias of every device in a stack,

$$I_{leak} \propto e^{(V_{GS} - V_{th})/nV_T} (1 - e^{-V_{DS}/V_T}),$$

so a NAND2 leaks differently in each of its four input states (stack effect). CharLib does the canonical thing: one DC operating point per input combination, all 2^N of them, with

$$P_{leak}(\text{state}) = V_{DD} |I_{VDD}(\text{state})|$$

read from the supply branch (leakage_power.py:67, 86–91), emitting one conditioned leakage_power { when : ... } group per state. lctime's characterize_leakage_power is an empty stub, never called — no leakage data at all. For the sequential cells (which CharLib's combinational-branch leakage never reaches, and which have no unique DC operating point anyway), the project's seq_leakage.py measures a *transient average* instead: a reset pulse forces a known state, then $P = V_{DD} |\overline{i_{VDD}}|$ with the mean taken by .meas tran AVG over the settled window 40–60 ns — a documented single-state approximation.

6 Sequential constraints

6.1 The physics: pushout

As the data edge approaches the clock edge, a flip-flop's internal regeneration starts from an ever-smaller initial imbalance and the clock-to-Q delay diverges logarithmically (regeneration time constant τ_m of the cross-coupled pair):

$$t_{C2Q}(t_{su}) \approx t_{C2Q, \infty} + \tau_m \ln \frac{t_0}{t_{su} - t_{su}^*}.$$

A setup (hold) constraint is therefore not a hard edge but a chosen point on this curve, and every characterizer must pick a *criterion*: how much delay pushout $\Delta d = t_{C2Q} - t_{C2Q, \infty}$ is tolerable. Inverting the curve shows how the criterion maps to the constraint:

$$t_{su}(\Delta d) = t_{su}^* + t_0 e^{-\Delta d / \tau_m},$$

so a tighter (smaller) Δd yields a larger, more conservative setup time — but only logarithmically slowly, which is why differing criteria still produce comparable libraries.

6.2 CharLib: what is stock and what is local

Stock CharLib 2.1.0 is largely unimplemented here: the sequential *delay* procedure returns immediately (delay.py:14–15, a # TODO stub), as do recovery, removal, min-pulse-width and the metastability binary search. Its one working constraint procedure is the C2Q *contour* method (per the setup/hold-interdependence literature): reference C2Q at a relaxed corner, validity criterion $t_{C2Q} \leq 1.2 \times t_{C2Q, ref}$, exponential bracketing plus binary search for the window edges, then an N×N sweep of the (setup, hold) rectangle and knee-point selection on the pass contour. On these HV cells the contour harness built transients that exhausted

memory, so the shipped library uses the project's procedures (`seq_delay_procedure.py`, registered through CharLib's own registry):

- **clock→Q**: preload pulse, data switch mid-period, measured clock edge at the grid slew; ngspice `.meas` from the clock's 50 % crossing (second rise) to the output's 50 % crossing, 20–80 % output transition; tables indexed clock-slew × load.
- **setup/hold**: pass/fail **bisection** on a two-pulse harness, search range $-2...+8$ ns, tolerance 10 ps. The pass criterion is deliberately two-part: the output must reach the target final state **and** its first half-rail crossing must occur within $1.5 \times$ the nominal C2Q ($\Delta d = 0.5 t_{C2Q, \infty}$, a *relative* degradation criterion) — final-state-only acceptance would silently ride through metastability. Latches are constrained against the closing enable edge; non-convergence counts as fail.

6.3 lctime

lctime's sequential path is genuinely implemented and more mathematically polished: it measures the pushout curve itself. `find_min_data_delay` doubles the setup/hold window until t_{C2Q} converges to $t_{C2Q, \infty}$ (abstol 1 ps); the constraint is then the root of

$$t_{C2Q}(t_{su}) - (t_{C2Q, \infty} + \Delta d) = 0,$$

with $\Delta d = 10$ ps (the default `max_pushout_time`), found with Brent's method (`optimize.brentq`, `xtol = 1` fs) after exponential bracketing — an *absolute* pushout criterion, in contrast to CharLib-local's relative one. It then re-solves setup with hold fixed at its unconditional minimum plus a 10 ps margin (and vice versa), which is what populates the constraint tables. Min-pulse-width uses the same root-finding on a two-pulse harness. Caveats visible in the source: the constraint search runs at **zero output load** (flagged # TODO in `flipflop.py`:196), `clk→Q` tables are taken at a fixed generous `setup = hold = 1` ns rather than the measured constraints, and **latches are not supported at all** (`main_lctime.py` aborts) — where the shipped library needed `en→Q` and closing-edge constraints for its five latches.

For this library's C2Q of a few hundred ps, CharLib-local's $\Delta d = 0.5 t_{C2Q, \infty}$ ($\sim 100+$ ps) versus lctime's 10 ps sit far apart on the pushout curve, yet by the logarithmic inversion above the extracted constraints differ only by $\tau_m \ln(\Delta d_1/\Delta d_2)$ — a few regeneration time constants, i.e. tens of ps. The bigger structural difference is coverage: only the local CharLib procedures produced latch data at realistic loads.

7 Numerics and simulator coupling

All three tools run the same engine — ngspice with the IHP OSDI-compiled PSP103 Verilog-A MOS models — through three different couplings, and two of the project's hardest-won findings are couplings, not physics:

	fo4.py	CharLib	lctime
coupling	ngspice -b batch		

	fo4.py	CharLib	lctime
		libngspice in-process (PySpice fork)	subprocess, ASCII wrdata
.meas support	native	fork-added; lost in subprocess mode	not used
integration method	trapezoidal (default)	trapezoidal, trtol = 1	trapezoidal (no options set)
timestep	20 ps fixed print step	slew/8 (delay), slew/ 10 (cap)	10 ps default, stop when breakpoints
stop condition	fixed 3 periods	autostop on .meas completion	breakpoint at 1 %/99 % V _{DD}
OSDI model loading	.spiceinit (batch honors it)	.spiceinit copied into run dir	.spiceinit copied by compare script
temperature	deck value	25 °C	hard-coded 25 °C (liberty header ignored)

The trapezoidal rule's local truncation error per step, $\epsilon \approx \frac{h^3}{12} |d^3v/dt^3|$, is what ngspice's timestep control bounds; trtol scales the tolerance it is compared against, so CharLib's trtol = 1 (versus the default 7) forces $\approx 7^{1/3} \approx 1.9 \times$ finer steps through the transitions — a deliberate accuracy/runtime trade in the delay decks.

The two coupling landmines, both documented in run_charlib.sh and the config generator because both produce *silently* wrong libraries: (1) with the ngspice-subprocess backend PySpice parses only the rawfile and never reads measurement results back, so every delay table comes back empty while leakage and capacitance still populate — the file looks plausible until a timing tool finds no arcs; shared mode is mandatory. (2) ngspice reads .spiceinit in batch and interactive mode but *not* in server mode (ngspice -s), so PySpice-driven runs never load the OSDI PSP103 models without the project's pre_osdi shim. lctime's file-based batch backend is immune to both by construction — its weakness is instead that it sets no simulator options at all and inherits whatever .spiceinit the working directory supplies, and that it stamps .option TEMP=25 regardless of the library header (the nom_temperature it parses is never propagated; characterizing a non-25 °C corner silently produces 25 °C data).

8 Can lctime use the charge-integration method?

As shipped in 0.0.26 — no. The verdict rests on three code facts:

1. The input-capacitance procedure is irreducibly the constant-current slope method: it drives the pin from a current source, records *only* the pin voltage (output_currents=[], input_capacitance.py:253–257), and computes $C = I \Delta t / \Delta V$ from two threshold crossings (lines 273–297). There is no charge in sight.

2. No code path in the package integrates a current: a search over the installed tree finds no `trapz/cumsum` and no `ngspice integ` in any generated control block.
3. The 10 μA source magnitude is a hard-coded constant (`util.py:162`), not reachable from the CLI — so even the existing method cannot be tuned, let alone replaced, by configuration.

But the port is small, because the data path already exists. For its power tables `lctime` already requests branch currents through named voltage sources and gets them back aligned with the time vector — both backends implement it (`wrdata ... i(vpin)` file-based; `print i(vpin)` interactive) — and the combinational routine already retrieves the *gate* current `-currents["V<pin>"]` for its energy term. Charge-integration capacitance is the same measurement pointed at a different source. The patch, confined to `characterize_input_capacitances()`:

1. **Drive voltage, not current:** declare the active pin as a voltage input and replace the `I<pin>` PWL current source with the `StepWave` ramp the delay code already uses (using the Liberty-correct $T = s/0.6$ stretch while at it).
2. **Ask for the current:** `output_currents=["V<pin>"]` instead of the empty list.
3. **Integrate:** replace the secant-slope block with

$$Q = \int_{\text{edge}} i_{\text{pin}}(t) dt \approx \text{np.trapz}(-i_{\text{Vpin}}, t), \quad C = \frac{|Q|}{V_{DD}},$$

using the trapezoidal sum (not the rectangle rule the power code uses) since the time grid is adaptive.

Nothing else in the harness changes — grids, state enumeration, reduction, and Liberty emission all operate on the returned scalar. On the evidence of this library, the payoff is the difference between +52 % and +9.7 % on `inv_1`'s pin capacitance, i.e. between a wire-load estimate that is wrong by half and one within the Miller-charge ambiguity that any single-number capacitance carries. (`lctime` is AGPL-3.0-or-later; a modified copy used in-house carries no distribution obligation, and an upstream contribution would resolve the license question entirely while fixing the rise/fall label swap noted in §3.)

9 Provenance of the shipped Liberty: no `lctime` data

A natural question given three tools: which of them actually populated `lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib`? The answer is that every number in the shipped file traces to the `CharLib` flow, and **none to `lctime`**:

Liberty content	produced by
combinational <code>cell_rise/fall</code> , <code>rise/fall_transition</code> tables	<code>CharLib 2.1.0</code> (<code>run_charlib.sh</code>), post-processed by <code>fix_lib.py</code>
pin capacitance / <code>rise_capacitance</code> / <code>fall_capacitance</code>	<code>CharLib</code> charge_integration procedure

Liberty content	produced by
per-state leakage_power groups (combinational)	CharLib all-states DC enumeration
clk→Q and en→Q arcs, setup/hold constraint tables	local seq_delay_procedure.py inside CharLib's registry, merged by merge_lib.py, fixed by fix_lib_seq.py
sequential and tie-cell cell_leakage_power	local seq_leakage.py / tie_leakage.py (transient average)
slew and load index grids	thin-oxide grids × the fo4.py ratios (2.66 delay, 2.20 capacitance)
functions, pin directions, ff/latch groups	lifted from the thin-oxide Liberty (logic unchanged by the transform)

lctime's role was strictly read-only verification: lctime_compare.py re-characterizes eight combinational cells on the same grids and models and compares its output *against* the shipped tables — the 2.9 % median delay agreement, the −19 % slew-convention band and the +52 % pin-capacitance finding of the preceding sections are all products of that comparison, and none of its data flows back. Given what the comparison revealed about the slope-method capacitance (§3) and the slew convention (§4), that separation is the correct engineering outcome, not an accident of history. A text search of the shipped file confirms it carries no lctime or LibreCell traces.

10 Conclusions

- All three input-capacitance methods are averages of the same $C_{in}(v)$ over different windows**, and their results order exactly as the windows predict: rail end-points 5.87 fF < full-swing charge mean 6.44 fF < Miller-peak window mean 8.92 fF. Charge integration is the defensible production choice — it measures the charge a driver actually delivers — with the rail-biased AC value as the clean lower anchor.
- CharLib is Liberty-correct on the slew convention; lctime is not.** The $T = s/0.6$ ramp stretch is the single largest systematic between the two characterizers (−19 % at multi-ns slews, mechanism confirmed by re-interpolation at 0.6 s to 0.25 %); where the convention doesn't bite, the tools agree to a median 2.9 % on delays and 0.0 % on transitions.
- Neither tool covers the full Liberty power model.** CharLib 2.1.0 has no internal-power procedure at all; lctime writes power tables but with a biased quadrature and without subtracting the $C_L V_{DD}^2$ load term, so they double-count against STA switching power. Leakage is CharLib-only (all-states DC enumeration; the project adds transient single-state leakage for sequential cells).
- Sequential characterization shipped only because of local code:** upstream CharLib's sequential delay is a stub and its contour method was unusable here; the local bisection (relative 1.5× C2Q criterion) and lctime's Brent-on-pushout (absolute 10

ps) are both principled points on the same metastability curve, differing mainly in coverage — lctime cannot do latches and constrains at zero load.

5. **lctime can adopt CharLib’s charge-integration method with a three-part, single-function patch** — the branch-current infrastructure it needs is already exercised by its own power measurement. Until then, its pin capacitances should not be used for anything load-sensitive.

11 Appendix: provenance

item	identity
reference deck	work/fo4.py, 96 lines, ngspice batch
CharLib	2.1.0, git stineje/CharLib commit 6859faf, venv /foss/tools/charlib, install hash-verified unmodified
PySpice	1.6 fork, git infinitymdm/PySpice commit da81c4d (adds .meas readback)
lctime	0.0.26, /usr/local/lib/python3.12/dist-packages/lctime
models	IHP SG13G2 PSP103 Verilog-A via OSDI, cornerM0Shv.lib mos_tt, 3.3 V, 25 °C
shipped library	lib/ sg13g2_stdcell_hv_typ_3p30V_25C.lib, thresholds 20/80/50, 7×7 NLDM grids
cross-check	work/lctime_compare.py: 8 cells, 3 132 aligned points
local CharLib adaptations	charlib_patched.py (case-insensitive branch lookup, procedure registration), seq_delay_procedure.py (clk→Q, setup/hold bisection), seq_leakage.py, gen_charlib_config.py (grids ×2.66/×2.20, charge integration selected), fix_lib.py / fix_lib_seq.py (header and sequential emission repairs)

Key source locations cited: CharLib procedures/combinational/delay.py (stimulus 75–83, measurements 101–112, worst-case 188–190), pin_capacitance/charge_integration.py (integration 100–104, formula 124–135), combinational/leakage_power.py (67, 86–91); lctime characterization/timing_combinatorial.py (StepWave 186–191, energy 263–274), characterization/input_capacitance.py (current source 235–243, slope 273–297, label swap 318–321), characterization/timing_sequential.py (pushout root-finding 1018–1228), ngspice_subprocess.py (deck 122–187, temperature 83).